

Question

Speculate on the structures and reaction mechanisms of **E** and **F** in Figure 1.

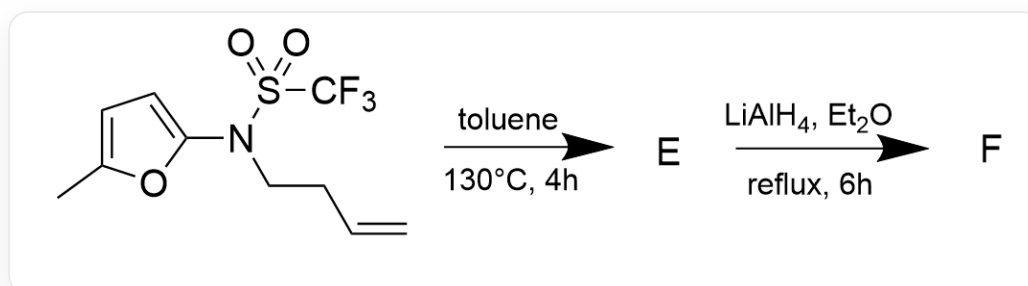


Fig. 1, the figure shows a two-step reaction. The first step is described by SMILES as: CC1=CC=C(N(S(=O)(=O)C(F)(F)F)CCC=C)O1>>[[E]], reaction conditions are toluene, 130°C, 4h. The second step is described by SMILES as: [[E]]>>[[F]], reaction conditions are LiAlH_4 , Et_2O , reflux, 4h.

The following statements are made:

- E** contains two rings of six members or less.
- The formation of **E** involves an intermediate containing 3 five-membered rings.
- The mechanism for the formation of **E** is an ionic mechanism.
- F** can eliminate one molecule of hydrogen gas to form a stable conjugated system.

Which of the following options contains all correct statements and the greatest number of correct statements:

A. All other options are incorrect

B. 1

C. 2

D. 3

E. 4

F. 1,2

G. 1,3

H. 1,4

I. 2,3

J. 2,4

K. 3,4

L. 1,2,3

M. 1,2,4

N. 1,3,4

O. 2,3,4

P. 1,2,3,4

Answer

Correct Answer: **M**

Detailed Explanation

Step one: The high temperature represents typical conditions for a pericyclic reaction. Observing the molecular structure, it can be found that the furan ring can act as a diene, while the terminal double bond can act as a dienophile. The two undergo a D-A reaction to generate the polycyclic intermediate shown in Figure 2:

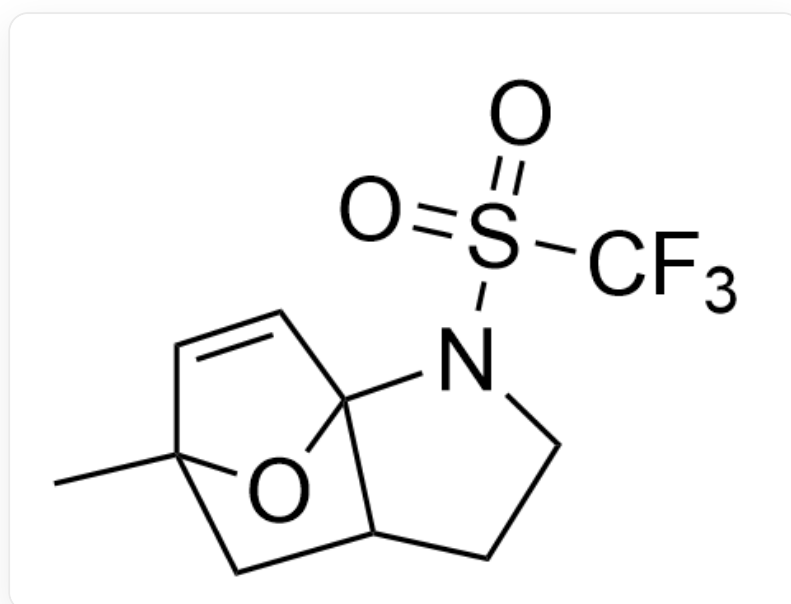


Fig. 2, the molecule in the figure is described by SMILES as: CC1(C=C2)OC32N(S(=O)(C(F)(F)F)=O)CCC3C1

CHECKPOINT

1 PTS

D-A reaction generates the intermediate shown in Figure 2:

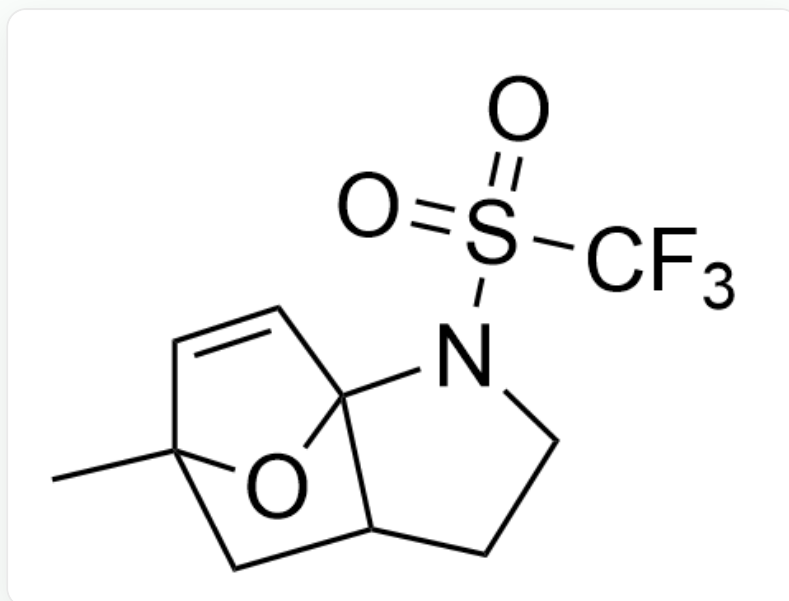


Fig. 2, the molecule in the figure is described by SMILES as: CC1(C=C2)OC32N(S(=O)(C(F)(F)F)=O)CCC3C1

This intermediate can further open the bridged ring structure and eliminate one molecule of water to form a benzene ring structure, which is the structure of **E**, as shown in Figure 3:

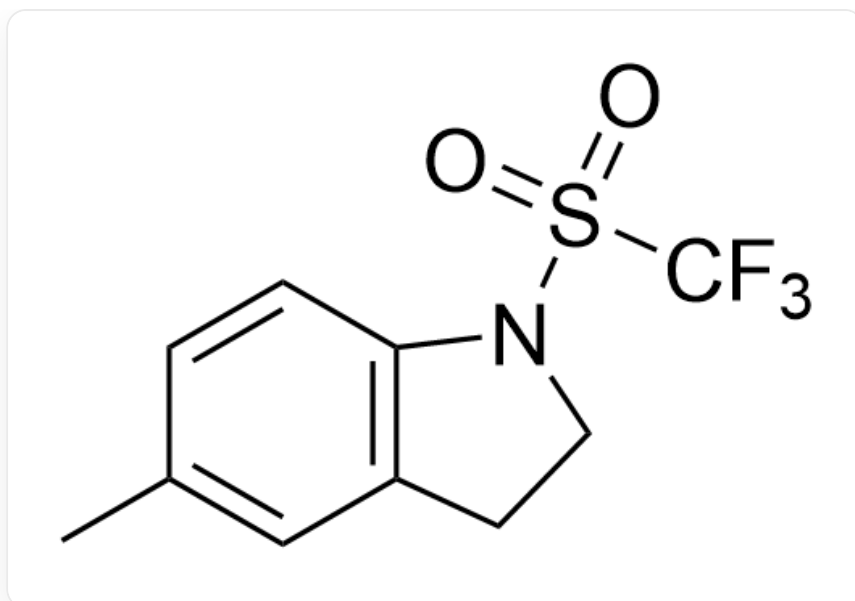


Fig. 3, the molecule in the figure is described by SMILES as: O=S(N1CCC2=C1C=CC(C)=C2)(C(F)(F)F)=O

CHECKPOINT

1 PTS

Open the bridged ring structure and eliminate water to form a benzene ring. The structure of **E** is shown in Figure 3:

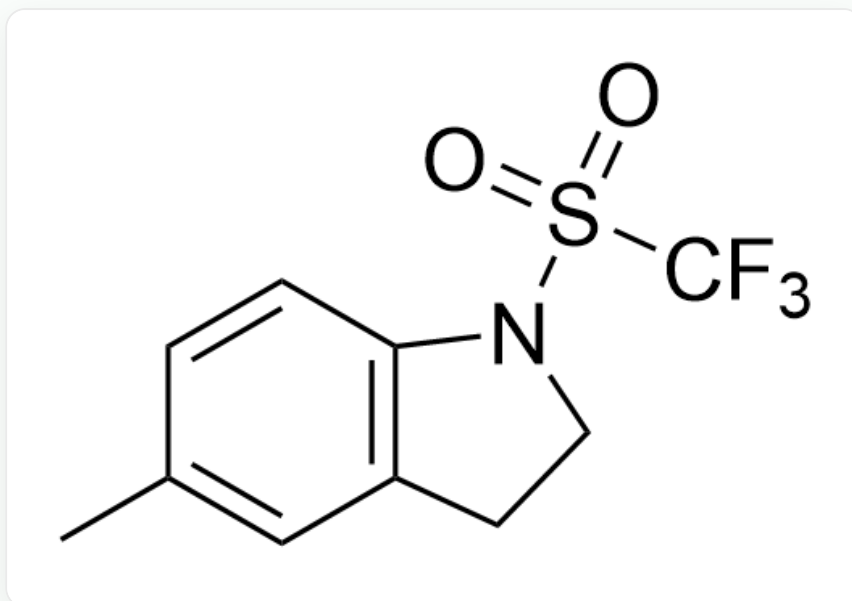


Fig. 3, the molecule in the figure is described by SMILES as: O=S(N1CCC2=C1C=CC(C)=C2)(C(F)(F)F)=O

E contains one six-membered ring and one five-membered ring, so statement 1 is correct.

The formation of **E** involves the intermediate shown in Figure 2, which contains three five-membered rings (and one six-membered ring), so statement 2 is correct. The formation of **E** is a D-A reaction with a pericyclic mechanism, so statement 3 is incorrect.

E is reduced by lithium aluminum hydride to remove the trifluoromethanesulfonyl group, yielding **F**, with the structure shown in Figure 4:

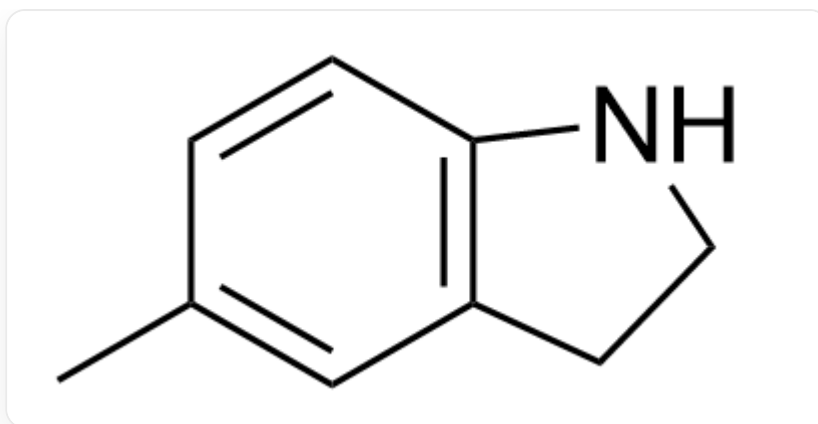


Fig. 4, the molecule in the figure is described by SMILES as: CC1=CC2=C(C=C1)NCC2

CHECKPOINT

1 PTS

The trifluoromethanesulfonyl group is reduced by lithium aluminum hydride to give **F** as shown in Figure 4:

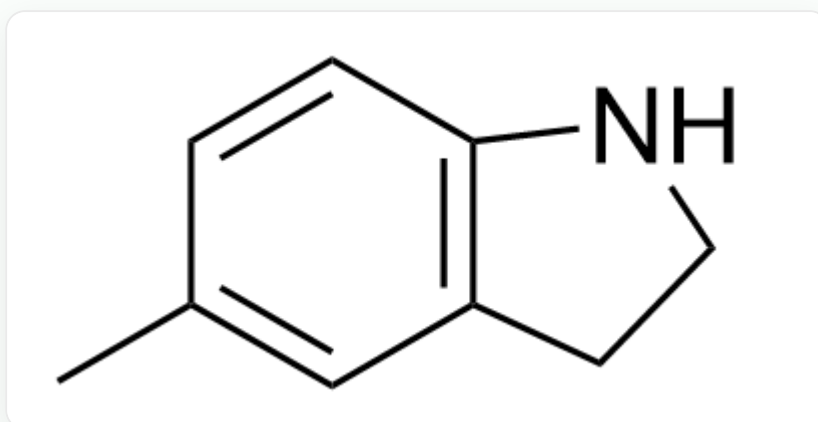


Fig. 4, the molecule in the figure is described by SMILES as: CC1=CC2=C(C=C1)NCC2

F can eliminate one molecule of hydrogen to form an aromatic indole ring, so statement 4 is correct.